

HAT-P-12b

R-0755

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_240401 · created 2026-07-28T03:26:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460401.9043 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.052
Transit depth (Rp/Rs)^2	1.81 +/- 1.39 [%]
Orbital Inclination (inc)	87.15 +/- 1.69
Airmass coefficient 1 (a1)	0.645 +/- 0.074
Airmass coefficient 2 (a2)	0.37 +/- 0.16
Scatter in the residuals of the lightcurve fit is	10.94 %
Best Comparison Star	#3 - [605, 338]
Optimal Method	PSF photometry
Transit Duration (day)	0.076 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.134 ± 0.052 vs 0.1406 (deviation 0.12σ)
Duration fit vs published	0.076 d vs 0.0975 d (deviation 0.81σ)
Mid-time O-C	-8.1 min
Depth SNR	2.5
Residual scatter	10.94 %

Light curve

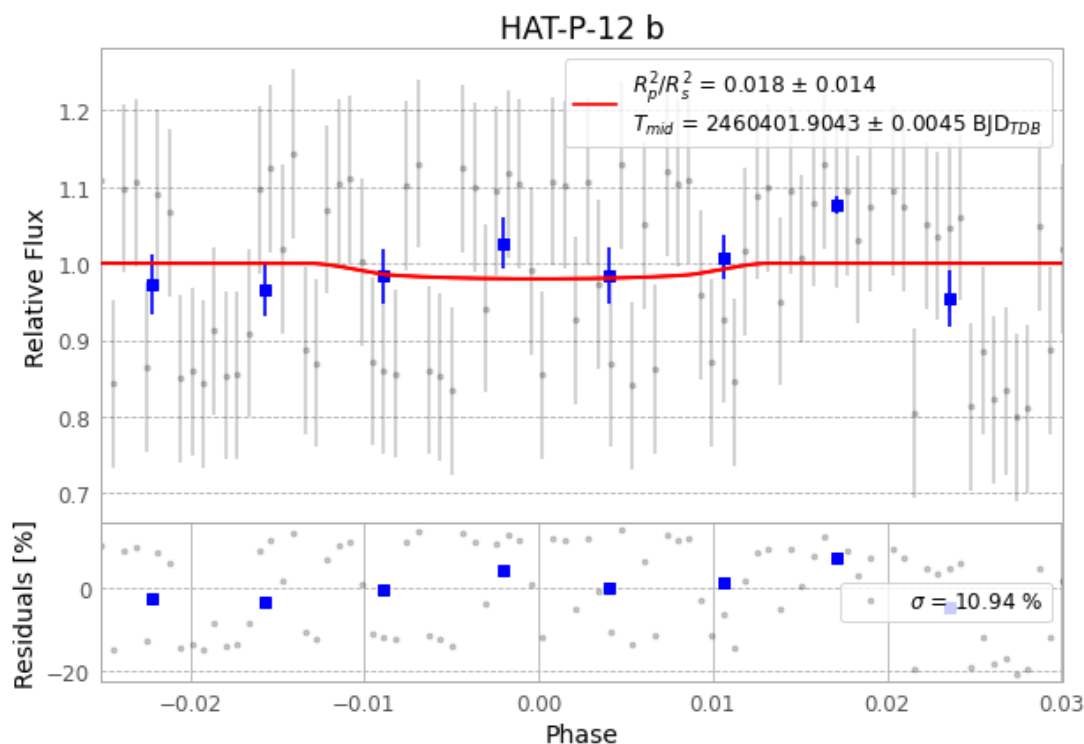


Figure 1 — FinalLightCurve_HAT-P-12 b_2024-04-01.png (SHA-256 5b384a244e644c4c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 457], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 67e4ffe680bff15a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

86 FITS frames (56 MB), HATP-12240401073915.FITS → HATP-12240401115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-240401.log (89d1c2ad97af...), FinalLightCurve_HAT-P-12 b_2024-04-01.png (5b384a244e64...), FinalLightCurve_HAT-P-12 b_2024-04-01.csv (031e81c7db60...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 03:26Z.